QQ-T-191C
January 6, 1972
SUPERSEDING
Fed. Spec. QQ-T-191B
December 2, 1964

FEDERAL SPECIFICATION

TERNE SHEETS (LONG AND SHORT TERNE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers steel sheet coated with lead-tin alloy. Long terne sheet is supplied in both coils and cut lengths. Short terne is supplied in cut lengths only.

1.2 Classification.

1.2.1 Types. Short terne sheet covered by this specification shall be furnished in the following types according to ductility, as specified in the contract or order.

Type I - Commercial quality
Type II - Drawing quality
Drawing quality, special killed

1.3 Coating weights. The weight of coating on terne sheet is the total amount on both surfaces of the sheet. Long terne is expressed in ounces per square foot of sheet in accordance with designations shown in ASTM A 308. The weight of coating for short terne is expressed in pounds per double base box. Short terne sheet covered by this specification shall be furnished in the following coating designations as specified in the contract or order:

SCMT (Special Coated Manufacturing Terne)
Manufacturing ternes (nominal 6 pounds per double base box)
8 pounds (nominal 8 pounds per double base box)

FSC 9515

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1.4 Grades. Short terne sheet covered by this specification shall be furnished in the following grades, as specified in the contract or order:

Grade A - Primes with seconds arising
Grade B - Unassorted

1.5 Finishes. Terne sheet covered by this specification shall be furnished in commercial finish.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard:

Fed. Test Method Std. No. 151 - Metals; Test Methods

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from the established distribution points in their agencies.)

Military Standards:

MIL-STD-163 - Steel Mill Products, Preparation for Shipment and Storage
MIL-STD-129 - Marking for Shipment and Storage

(Copies of Military Specification and Standard required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

A 308 - Specification for Long Terne Iron or Steel Sheets, Coils, and Cut Lengths.
A 309 - Weight and Composition of Coating on Long Terne Sheets by the Triple Spot Test.
A 568 - General Requirements for Carbon and High-Strength Low Alloy Steel, Hot-Rolled Strip, Hot-Rolled Sheets, and Cold-Rolled Sheets.
A 623 - Specification for General Requirements for Tin Mill Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa., 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material

3.1.1 Long terne sheets shall meet all the requirements of ASTM A 308.

3.1.2 The raw materials used to produce short terne sheet shall be such as to meet the requirements of this specification. The base metal used for the short terne sheet shall be made by the basic open hearth process, electric furnace process, or basic oxygen process.

3.2 Chemical requirements

3.2.1 Base metal. When specified in the contract or order, the base metal shall be copper bearing, low carbon steel. Copper content shall be not less than 0.20 percent by cast analysis and 0.18 percent by product analysis. If stainless steel is ordered, type 304 will be furnished unless otherwise specified (see 6.2).

3.2.2 Terne coating. The short terne metal used shall have a minimum of 15 percent tin and a minimum of 99 percent tin plus lead, except that in the case of special coated manufacturing terne (SCMT), the minimum tin requirements shall not apply.

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3.2.2.1 When specified in the contract or order, the contractor shall furnish an analysis showing the percentages of the elements designated in 3.2.2.

3.3 Weight of coating

3.3.1 Short terne. The weight of the terne coating for short terne shall be not less than 70 percent of the ordered weight for any single plate when determined by the triple spot test.

3.4 Lot weight

3.4.1 Short terne. The weight of a lot of short terne shall be as calculated from the amounts shown in table I within the permissible variations shown in table II.

TABLE I. Approximate weight of coated short terne

Base weight[1/] per single base box (uncoated) sheets)	Approximate weight of coated sheets per single base box		
	SCMT (Special coated manufacturing terne)	Manufacturing terne	8-pound coating weight
Pounds	Pounds	Pounds	Pounds
80	80	80	80
85	85	85	85
90	90	90	90
95	95	95	95
100	100	100	100
107	107	107	107
112	112	112	112
118	118	118	118
128	128	128	128
135	135	135	135
139	-	139	139
148	-	148	148
155	-	155	155
168	-	168	168
175	-	175	175
180	-	180	180
188	-	188	188
195	-	195	195

[1/] Base weights heavier than 195 lbs in manufacturing ternes and 8-pound ternes may be obtained by negotiation between purchaser and contractor.

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TABLE II. Permissible variation in lot weight for short terne

Quantity	Permissible variations, in percent of lot weight	
	plus percent	minus percent
Individual plates	10	10
1 to 15 packages, inclusive	6	4
Over 15 to 200 packages, inclusive	4	4
Over 200 packages	2-1/2	2-1/2

3.5 Dimensional tolerances. Tolerances for short terne sheets shall be in accordance with the following ASTM methods as applicable.

	Short terne 1/
Weight tolerances	A 623
Width and length tolerances	A 623
Out of square tolerances	A 623
Resquare tolerances	2/
Camber tolerances	A 623
Length and width (stretcher leveled)	- -
Flatness	- -
Flatness (stretcher leveled)	3/

1/ Short terne tolerances shall be the same as for tin mill products according to ASTM A 623.
2/ Not available as resquared sheet.
3/ Flat tolerances, short terne. Maximum deviation from a horizontal flat surface shall be 1/8 inch.

3.6 Bend test. All terne sheets shall be capable of being bent at room temperature through an angle of 180° flat on itself in any direction without flaking of the coating or cracking of the base metal.

3.7 Grade.

3.7.1 Grade A-primes with seconds arising. Primes with seconds arising shall consist of sheets as defined in 3.9.2 and 3.9.3. The seconds shall be packaged separately and shall not exceed 20 percent of the quantity ordered.

3.7.2 Grade B-unassorted. Unassorted sheets shall consist of primes and seconds as defined in 3.9.2 and 3.9.3. Seconds need not be packaged separately.

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3.8 Drawing quality terne sheets. Terne sheets of drawing quality shall be suitable for such drawing and forming operations as are specified in the contract or purchase order, within the limits of good cold drawing practice, and shall pass such drawing and forming tests as are specified in the contract or order.

3.9 Workmanship.

3.9.1 Base metal. The base metal shall be free from laminations, blisters, slivers, open seams, pits, ragged edges, holes, turned down corners, and internal or surface defects which will impair the intended use.

3.9.2 Primes. Primes sheets, in addition to being free from the defects listed in 3.9.1, shall be free from bare or imperfectly coated spots readily observable by the unaided eye, deep scratches, heavy list edges, and other defects visible to the unaided eye.

3.9.3 Seconds. Seconds may contain minor imperfections in coating, base metal or other manufacturing defects, such as may occur in good mill practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot. Unless otherwise specified in the contract or order, a lot shall consist of all material of the same class, base weight or thickness, coating designations, grade, nominal base metal composition, and size submitted for inspection at one time.

4.3 Sampling.

4.3.1 For coating weight test. One short terne sheet from each 50 packages, or fraction thereof, shall be selected for testing.

4.3.2 For sheet weight test (short terne). From one to ten coated sheets from each 50 packages or fraction thereof shall be selected for the sheet weight test.

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4.3.3 Composition of terne metal. Determination of tin and lead content shall be made by the manufacturer periodically and certification of analysis shall be furnished upon request to the procuring agency.

4.3.4 Bend test samples. When specified in the contract or order, at least two samples from each sheet selected in accordance with 4.3.1, shall be taken for bend tests.

4.4 Examination.

4.4.1 Visual. Unless otherwise specified in the contract or order, all terne sheets shall be subject to visual examination to determine compliance with the requirements for grade (see 3.7) and workmanship (see 3.9).

4.4.2 Dimensional. All short terne sheets shall be subject to examination to determine compliance with the dimensional requirements (see 3.5). Straightness and squareness shall be determined as specified in 4.4.2.1 and 4.4.2.2, respectively.

4.4.2.1 Straightness. Straightness shall be determined by placing the sheet on a level surface so that the arc or departure from straightness is horizontal. The maximum depth of arc shall be measured to the nearest 1/32 inch by means of a straight edge and a steel scale.

4.4.2.2 Squareness. Squareness shall be determined by placing a straight edge at right angles to the side of the sheet, touching one corner, and measuring the deviation at the opposite corner at the same time.

4.4.3 Preparation for shipment. Examination of the packing and marking for shipment shall be made for compliance with the requirements of section 5.

4.5 Tests (for short terne sheets).

4.5.1 Unless otherwise agreed upon between the producer and the procuring activity, the coating weight test shall be conducted as shown in ASTM A 309.

4.5.1.1 Sheets coated in cut lengths. From each sheet taken in accordance with 4.3.1, one specimen shall be cut from the center of the sheet and the other two from diagonally opposite corners, adjacent to and within the boundary formed by lines not less than 2 inches from the sides and 4 inches from the ends.

4.5.2 Weight and composition. The weight and composition of terne coating for short terne shall be determined in accordance with method 514 of Fed. Test Method Std. No. 151.

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4.5.3 Bend test. When specified in the contract or order, at least two bend test specimens shall be cut from each sample sheet (see 4.3.4). The specimens shall be 2 inches wide with a minimum length of 6 inches. The specimens shall be bent at room temperature through an angle of 180° flat on itself in any direction and then examined for conformance to 3.6.

4.6 Rejection and retest. Unless otherwise specified in the contract or order, rejection and retest shall be conducted in accordance with the general section of Fed. Test Method Std. No. 151.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Cleaning, drying, preservation, and packaging shall be in accordance with MIL-STD-163.

5.1.2 Level C. Cleaning, drying, preservation, and packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A or level C as specified (see 6.2).

5.2.1 Level A. Packing shall be in accordance with MIL-STD-163.

5.2.2 Level C. Packing shall be in accordance with commercial practice adequate to insure acceptance and safe delivery by the carrier for the mode of transportation employed.

5.3 Marking. In addition to any special marking required by the contract or order, shipment shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Long terne sheets. Long terne sheets (type I) are for general use, such as cornices and calamein door covering for building construction. Type II long terne sheets are for special purposes where drawing or forming operations are too severe for type I terne.

6.1.2 Short terne. Short terne sheets are intended for use in the fabrication of metal containers and other sheet metal applications where corrosion resistance or subsequent hot-dipping requires a terne coating.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Type (see 1.2).
c. Coating designation (see 1.3)
d. Grade (see 1.4).
e. Thickness (if long terne) or base weight (if short terne).
f. When a statement of chemical analysis is required (see 3.2.2.1)
g. Lot size, if necessary (see 4.2).
h. When bend test is required (see 4.3.4 and 4.5.3).
i. When different visual examination sampling is required (see 4.4.1).
j. When different coating weight test is permissible (see 4.5.1)
k. When variation in rejection and retest requirements are permitted (see 4.6)
l. Preparation for delivery requirements (see section 5).
m. Maximum gross weight of container.

6.3 Coating weights.

6.3.1 Long terne coating designation LT-01 is intended for general use where the surface is to be painted or only temporary protection is required, such as liners for packing cases. A zinc coating may be applied to this material after forming, by the hot dip process, the terne coating acting as a flux. The remaining coating designations are intended for more permanent protection against corrosion as in the case of cornices for buildings. Materials with these designations may be formed in a quick acting brake without danger of flaking of the coating such as occurs with heavily zinc-coated sheets. Material with coating designations LT-85 and LT-110 are not ordinarily carried in warehouse stocks and should be ordered only when specially manufactured material is warranted.

6.3.2 Special Coated Manufacturing Terne, commonly abbreviated SCMT, is terne sheet with no specific amount of coating, but customarily carrying the lightest practical coating consistent with thorough coverage of the plate.

6.4 Unassorted terne consists of primes and seconds as they occur in the course of manufacture, without seconds being separately packaged. No maximum on the percentage of seconds is imposed as the amount cannot be determined without segregation. Where such a product can be used, it is a more economical grade than primes or primes with seconds arising.

6.5 Definitions.

6.5.1 Package. A "package" consists of 112 plates.

6.5.2 Single base box. A "single base box" consists of 112 plates, 14 by 20 inches (equal to 31,360 square inches of plate) or equivalent.

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6.5.3 Double base box. A "double base box" consists of 112 plates, 20 by 28 inches (equal to 62,720 square inches of plate) or equivalent.

6.6 Roofing terne. This specification is not intended to cover roofing terne. Terne plate for roofing is covered by QQ-T-201.

MILITARY CUSTODIANS:

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CIVIL AGENCY INTEREST:

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

QQ-T-191C, Terne Sheets (Long and Short Terne)

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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